THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The outermost layer of the Earth is the: 1

(a) crust **(b)** mantle **(c)** outer core **(d)** inner core

2. Reflection from a smooth mirror is: 1

(a) diffuse reflection **(b)** dispersion

(c) refraction **(d)** regular reflection

3. The time from one new moon to the next new moon is about: 1

(a) 29.5 days **(b)** 7 days **(c)** 24 hours **(d)** 365 days

4. Which of the following is a correct food chain? 1

(a) snake, frog, grass **(b)** grasshopper, snake, grass

(c) frog, grass, snake **(d)** grass, grasshopper, frog, snake

5. The percentage of oxygen in air is about: 1

(a) 21 per cent **(b)** 50 per cent **(c)** 78 per cent **(d)** 1 per cent

6. The mirror used as a rear-view mirror in vehicles is: 1

(a) cylindrical **(b)** concave **(c)** convex **(d)** plane

7. The calendar based on the phases of the Moon is the: 1

(a) daily **(b)** solar calendar **(c)** lunar calendar **(d)** weekly calendar

8. Which gas do green plants take in during photosynthesis? 1

(a) nitrogen **(b)** hydrogen **(c)** carbon dioxide **(d)** oxygen

9. Most of the Earth's fresh water is stored as: 1

(a) lakes **(b)** rivers

(c) ice caps and glaciers **(d)** clouds

10. A ray strikes a plane mirror at an angle of incidence of 35 degrees. The angle of reflection is: 1

(a) 35 degrees **(b)** 55 degrees **(c)** 145 degrees **(d)** 70 degrees

11. The Earth completes one revolution around the Sun in about: 1

(a) 30 days **(b)** 27 days

(c) 24 hours **(d)** 365 and a quarter days

12. Which of the following is a biotic component? 1

(a) bacteria **(b)** air **(c)** water **(d)** soil particles

13. The layer of the Earth between the crust and the core is the: 1

(a) atmosphere **(b)** crust **(c)** mantle **(d)** hydrosphere

14. We can see a non-luminous object because it: 1

(a) emits light **(b)** reflects light into our eyes

(c) is hot **(d)** absorbs all light

15. A moon phase when the full face of the Moon is visible is the: 1

(a) first quarter **(b)** full moon **(c)** crescent **(d)** new moon

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why is the shadow of a stick shortest at noon? 2

17. What do decomposers do? Name two. 2

18. Why is the Earth called a 'unique planet'? 2

19. Differentiate between convex and concave lenses (one point each). 2

20. Name the two types of calendars based on the Sun and on the Moon. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Why is the Earth's distance from the Sun important for life? 3

22. Mention three uses of lenses. 3

23. Explain what is meant by 'a day' as the time for one rotation, and a 'month' and a 'year' with astronomical references. 3

24. Explain the importance of decomposers in nature. 3

OR

State three differences between the Earth and the Moon regarding life. 3

25. A ray of light makes 40 degrees with the surface of a plane mirror. Find the angle of incidence and the angle of reflection. 3

OR

Explain how the length and direction of the shadow of a stick change during the day. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the lens types, their action on light, the names of the principal focus and the focal length with uses. 5

OR

Explain why we see the Moon from Earth in phases and not always the full disc, using the positions of the Sun, Earth and Moon. 5

27. Explain how the removal of one species affects an ecosystem. Give an example of vultures or tigers. 5

OR

Explain the importance of the ozone layer, the harmful gases and the Montreal Protocol. 5

END OF QUESTION PAPER **Total: 50 Marks**

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